The Local Housing-Market Effects of Act 22/60 Investor Purchases: Data and Empirical Design

August 2026

1 Research question

When a tax-incentivized migrant (an Act 22/60 "Individual Investor" decree holder) buys a residential property in Puerto Rico, what happens to the housing market immediately around that property — to prices, and to who transacts?

2 Data construction

2.1 Identifying investor properties

- **Decree holders.** Government lists of individual grantees: 3,181 (Act 22) and 2,672 (Act 60) names; 5,770 distinct first–last pairs pooled.
- **CRIM cadastral registry.** Name searches against the CRIM parcel layer (via its ArcGIS REST service) matched 26,673 unique parcels, each with full attributes, centroid coordinates, and the parcel's *most recent* sale (price, date, buyer, seller). CRIM stores only the latest transaction per parcel.
- **Karibe property registry.** Registry name-search results; cadastral numbers extracted from registral descriptions and resolved in CRIM provide a second, largely non-overlapping source.
- **Match confidence.** Name→parcel matching is noisy for common names. A name returning *exactly one* property is treated as high-confidence ("unique match"): 810 parcels via CRIM, 755 via Karibe (surname-validated, geocoded), union **1,306 parcels** (259 cross-validated in both). The event studies use this base.
- **Coverage.** High-confidence identification covers 22.5% of searched names (26.7% including a validated multi-match recovery tier), versus 49–61% declared homeownership in the Act 22 annual reports. The gap is dominated by LLC/trust purchases (invisible to name search), name-format failures, and registry availability; the identified sample skews toward single-property, personal-name buyers.

2.2 Events

An **event** is a dated investor purchase: unique-match parcels with a valid CRIM sale date, collapsing same-day, same-location purchases (condo units bought together) into one event, giving 1,256 events (1,200 in the decree era, ≥ 2012, used in estimation). Purchases cluster heavily: the median event has 25 other events within 1 km (158 are fully isolated), so contamination is measured per event and used in robustness.

2.3 Outcomes: nearby sales

For each event, every CRIM parcel within 1,000 m with a recorded sale: 3.47M sale$\times$event pairs over 328K distinct parcels, each carrying distance, sale price/date/parties, property characteristics (lot size, assessed structure value, condo-unit and vacant-land indicators), and 2020 census tract. *Rings:* 0–250 m (treated), 250–400 m (buffer, dropped), 400–1,000 m (control). *Filters:* nominal transfers ($<$ \$10k), implausible dates, and sales of investor-matched parcels are excluded. *Caveat:* CRIM keeps only each parcel's last sale, so earlier sales are censored by later resales; the within-window, near-versus-far design mitigates this, and Figure 5 tests it.

2.4 Placebo events

1,500 purchases by non-corporate *individual* buyers in the investor price band (\$285K–\$3.9M, the p25–p95 of investor purchases), located within 250 m of a real event site (where the sales pool has full coverage) but more than three years apart in time. Far rings are truncated at 750 m for both the placebo run and a matched real-event baseline.

2.5 Buyer/seller origin proxy

Party names are classified against the Census 2010 surnames file (`pcthispanic`), taking the maximum across name tokens (robust to CRIM's inconsistent name ordering; appropriate under the Puerto Rican two-surname convention). Names with `pcthispanic` ≤ 20 are "non-Hispanic-named" (mainland proxy), ≥ 70 "Hispanic-named" (local proxy); corporate, ambiguous, and unmatched names are left unclassified, and the raw score is retained for re-thresholding. *Validity:* 67% of investor purchases have non-Hispanic-named buyers versus 10% of surrounding market sales. *Caveat:* the proxy captures *name* origin — stateside Puerto Ricans and other Hispanic-named mainlanders classify as local, so mainland penetration and displacement effects are understated.

2.6 Summary statistics

Table 1 summarizes the estimation samples: the ring-design sale sample, the Red Atlas tract price panel, the HMDA purchase panel, and the 2010 PRCS baseline tract characteristics used for balance and robustness.

Table 1: **Summary Statistics** This table presents summary statistics for the paper's main samples. Panel A: market sales within 1km of an identified decree-era investor purchase (ring design estimation sample: gap ring, junk/nominal prices, and investor-parcel sales excluded; events 2012+); name-classification indicators are defined over sales whose parties' surnames classify. Panel B: Red Atlas tract×year panel (transaction-count-weighted annual mean prices); identified purchases reported over the 235 treated tracts. Panel C: HMDA tract×year panel of originated first-lien owner-occupied 1–4 family purchases, 2012–2024. Panel D: 2010 PRCS tract characteristics (2020 boundaries, population-weighted).

	Mean (1)	Std. Dev. (2)	P25 (3)	Median (4)	P75 (5)	Observations (6)
Panel A: Ring-design sales (≤1km of an event)						
Sale price ($000s)	1,872.3	61,984.0	116.6	240.0	475.0	971,720
Log sale price	12.35	1.27	11.67	12.39	13.07	971,720
1{non-Hispanic-named buyer}	0.169	0.375	0.000	0.000	0.000	660,570
1{Hisp. seller → non-Hisp. buyer}	0.115	0.320	0.000	0.000	0.000	500,309
1{non-Hisp. seller → Hisp. buyer}	0.097	0.295	0.000	0.000	0.000	500,309
Panel B: Tract price panel (Red Atlas)						
Tract annual mean price ($000s)	148.5	122.9	91.2	122.3	168.5	16,344
Tract annual transactions	25.5	24.0	11.0	19.0	33.0	16,344
Identified purchases (treated tracts)	5.1	13.7	1.0	1.0	3.0	235
Panel C: HMDA purchases (2012–2024)						
Hispanic-borrower purchases	11.6	12.3	4.0	8.0	15.0	11,350
Non-Hispanic-borrower purchases	0.195	0.819	0.000	0.000	0.000	11,350
Mean borrower income ($000s)	54.7	79.8	34.8	43.9	58.0	11,100
Panel D: 2010 PRCS tract characteristics						
Median household income ($)	20,242	10,759	13,704	17,102	23,700	875
Median home value ($)	119,534	54,345	88,700	104,150	134,675	874
Median gross rent ($)	461	181	352	436	541	863
Poverty rate	0.457	0.165	0.347	0.473	0.576	877
BA+ share (25+)	0.208	0.128	0.120	0.175	0.257	878
Renter share	0.291	0.155	0.187	0.251	0.358	877
Vacancy rate	0.162	0.088	0.105	0.147	0.199	877
Seasonal-home share	0.032	0.060	0.006	0.017	0.036	877
Born-in-mainland share	0.050	0.025	0.031	0.047	0.064	878
Population	4,020	2,008	2,634	3,990	5,320	936

3 Empirical design

3.1 Panel construction

Because CRIM records one (the last) sale per parcel, sales are aggregated into an event×ring×period panel. Let e index events, $r \in \{\text{near}, \text{far}\}$ rings, and t calendar periods (years in the robustness suite). The cell outcome is the mean log price among the N_{ert} transactions in cell (e, r) in period t:

$$\bar{y}_{ert} \; = \; \frac{1}{N_{ert}} \sum_{j \in (e,r,t)} \ln P_j, \tag{1}$$

where P_j is the sale price of transaction j. Treatment is absorbing and turns on for the near cell in the event period t_e:

$$D_{ert} \; = \; \mathbf{1}\{r = \text{near}\} \times \mathbf{1}\{t \geq t_e\}, \tag{2}$$

so far cells are never treated and serve as clean controls. In the tract-fixed-effects specification, $\ln P_j$ is first residualized on census tract dummies at the sale level $(\ln P_j - \hat{\gamma}_{\tau(j)})$ and the residual is aggregated instead; results are unchanged.

3.2 Local projections difference-in-differences (LP-DiD)

Following Dube, Girardi, Jordà and Taylor (2023), the dynamic effect at each horizon h is estimated from a separate long-difference regression on the stacked cell panel. For post-treatment horizons $h = 0, 1, \ldots, H$:

$$\bar{y}_{er,\,t+h} - \bar{y}_{er,\,t-1} \; = \; \beta_h^{\text{LP-DiD}} \, \Delta D_{ert} \; + \; \delta_t^h \; + \; \varepsilon_{ert}^h, \tag{3}$$

and for pre-treatment horizons $h = 2, \ldots, Q$ (with $h = 1$ the normalization):

$$\bar{y}_{er,\,t-h} - \bar{y}_{er,\,t-1} \; = \; \beta_{-h}^{\text{LP-DiD}} \, \Delta D_{ert} \; + \; \delta_t^h \; + \; \varepsilon_{ert}^h, \tag{4}$$

where $\Delta D_{ert} = D_{ert} - D_{er,t-1}$ indicates that the near cell of event e enters treatment in period t, and δ_t^h are horizon-specific calendar-period effects.

Each horizon-h regression is estimated on the restricted "clean" sample

$$\Big\{ (e, r, t) : \quad \underbrace{\Delta D_{ert} = 1}_{\text{newly treated near cells}} \quad \text{or} \quad \underbrace{D_{er,s} = 0 \;\; \forall s}_{\text{never-treated far cells}} \Big\}, \tag{5}$$

which is the `nevertreated` option: the counterfactual change $\bar{y}_{er,t+h} - \bar{y}_{er,t-1}$ for a newly treated near cell is identified from far cells over the same calendar window, and no already-treated observation is ever used as a control (the negative-weighting problem of two-way fixed effects under staggered timing cannot arise). $\beta_h^{\text{LP-DiD}}$ is a variance-weighted average of event-specific effects; standard errors are clustered by cell. The identifying assumption is within-micro-neighborhood parallel trends: absent the purchase, prices of homes transacting 0–250 m from the site would have evolved like those 400–1,000 m away, conditional on calendar-time effects — shocks common to the micro-area (Hurricane María, the 2020–21 in-migration wave) difference out by construction, and equations (4) provide the testable pre-trend restrictions $\beta_{-h} = 0$.

Two scope conditions follow from the ring structure. First, the near–far contrast differences out any effect *common to the full 1-km neighborhood*: the design identifies the hyper-local increment around a purchase site, not the neighborhood-level effect of aggregate investor inflows, which requires the complementary tract-level staggered design using the tract×month price panel (Section 5). Second, because purchases cluster at the enclave scale, other investor purchases in dense areas fall disproportionately inside the near ring itself; the estimated contrast there measures the cumulative local dose of investor activity rather than the effect of a single purchase (see Figure 5).

3.3 Robustness to the estimation approach

The cell-mean panel is a construction choice; the ring literature (Linden and Rockoff 2008; Campbell, Giglio and Pathak 2011) instead estimates on individual transactions: stacked sale-level regressions of log price on near×post (or near×event-time dummies) with event×ring and event×calendar-year fixed effects, standard errors clustered by event. The two implementations also weight differently — transactions versus events — so agreement is informative rather than mechanical. Re-estimating the baseline this way yields a pooled near×post effect of +7.09 log points (s.e. 1.58), and +6.72 (1.32) with hedonic controls (log lot size, sub-unit and vacant-land indicators), against the cell-level LP-DiD baseline of +6.95 (1.81); the event-study paths coincide horizon by horizon (Figure 1).

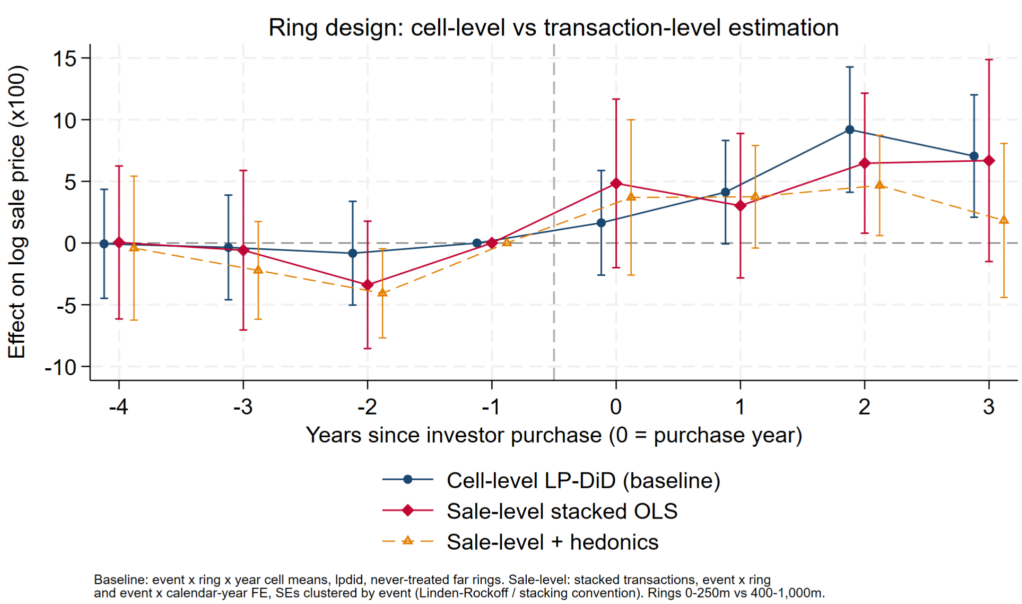

Figure 1: **Estimation-approach robustness.** The near-ring event study estimated three ways: the cell-level LP-DiD baseline; sale-level stacked OLS with event×ring and event×calendar-year fixed effects (the ring literature's convention), clustered by event; and the sale-level specification with hedonic controls. Rings 0–250 m vs 400–1,000 m. The three series coincide within sampling error at every horizon, and the agreement of transaction-weighted and event-weighted estimates indicates the pooled effect is not driven by a few high-volume events.

In the annual robustness suite $Q = 4$ and $H = 3$; the baseline is also estimated at quarterly ($Q = H = 8$) and half-yearly resolutions and with the Callaway–Sant'Anna (2021) group-time estimator $ATT(g,t) = \mathbb{E}[\bar{y}_t - \bar{y}_{g-1} \mid G_g = 1] - \mathbb{E}[\bar{y}_t - \bar{y}_{g-1} \mid \text{never treated}]$, aggregated to event time; results are similar across estimators and resolutions, and identical with and without tract fixed effects.

4 The eight figures

All figures plot LP-DiD event-study coefficients $\{\beta_{-Q}, \ldots, \beta_H\}$ from equations (3)–(4) at annual resolution (pre-window 4, post-window 3), with 95% confidence intervals, cells clustered. Period -1 is the omitted base; the dashed vertical line marks the treatment boundary.

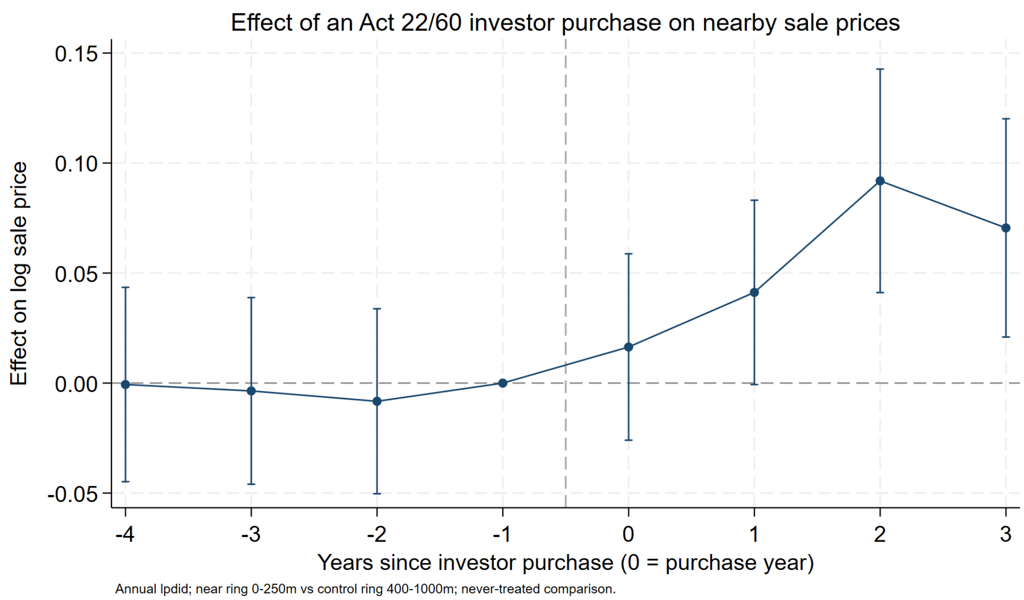

Figure 2: **Baseline event study.** Near-ring (0–250 m) versus far-ring (400–1,000 m) effects on log sale price by year relative to the investor purchase. The headline price bump; flat pre-period coefficients are the design's core credibility check, and the post-period path shows how quickly the premium appears and whether it persists.

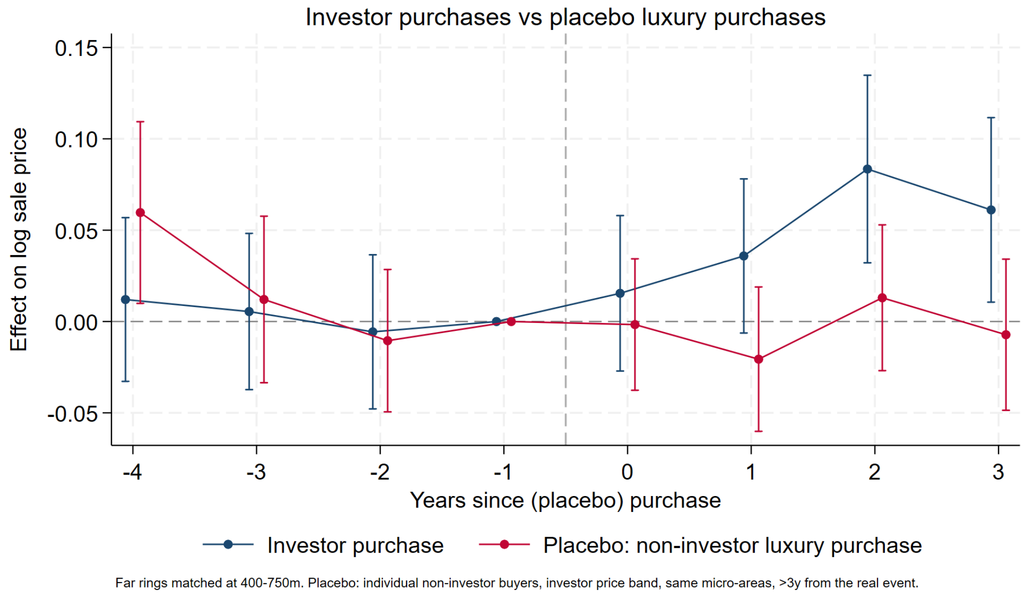

Figure 3: **Investor vs. placebo purchases.** The real event study overlaid with placebo events (non-investor individual buyers in the investor price band, same micro-areas, >3 years from the real event; far rings matched at 400–750 m). A flat placebo says nearby prices respond to *the investor*, not to any high-value transaction; because unmatched investors contaminate the placebo toward finding an effect, a null placebo is conservative evidence of investor specificity.

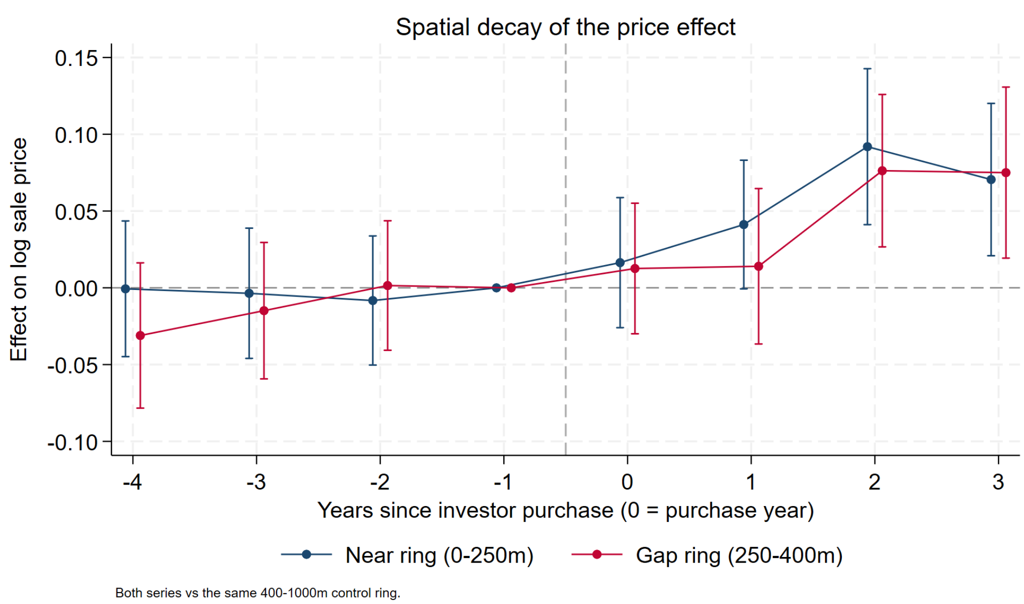

Figure 4: **Spatial gradient.** Near-ring (0–250 m) and gap-ring (250–400 m) series, each against the same far control. A genuine hyper-local spillover should decay with distance (near > gap > 0); near ≈ gap would instead indicate a neighborhood-wide shock partially shared by the control ring.

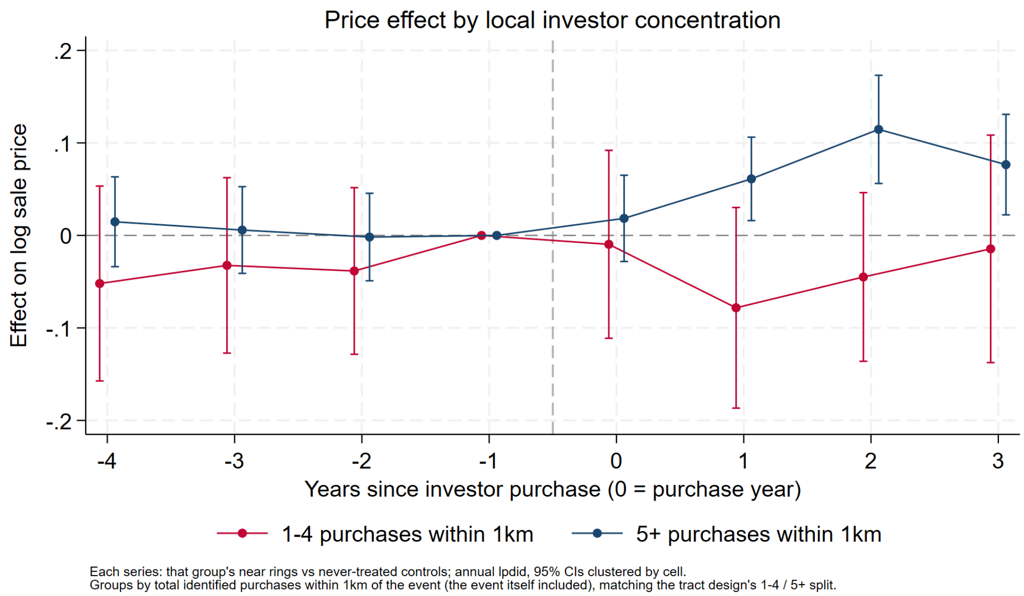

Figure 5: **Estimated contrast by local investor concentration.** Separate event studies for events with 1–4 vs. 5+ total identified purchases within 1 km (the event itself included), matching the tract design's dose split. Because purchases cluster at the enclave scale (condo buildings, resort communities), other events in dense areas fall disproportionately *inside* the near ring, so the near–far contrast there reflects cumulative local investor exposure rather than a single purchase. Low-concentration events show no price effect (post−pre −2.7, s.e. 3.5); 5+ areas show +6.4 (s.e. 1.5): single-purchase channels (comps entry, one renovation, signaling) appear weak on their own, and the price response is a phenomenon of investor *concentration* — the same 1–4/5+ pattern as the tract design. Pre-period coefficients are flat in both groups, inconsistent with investor waves selecting into already-appreciating pockets.

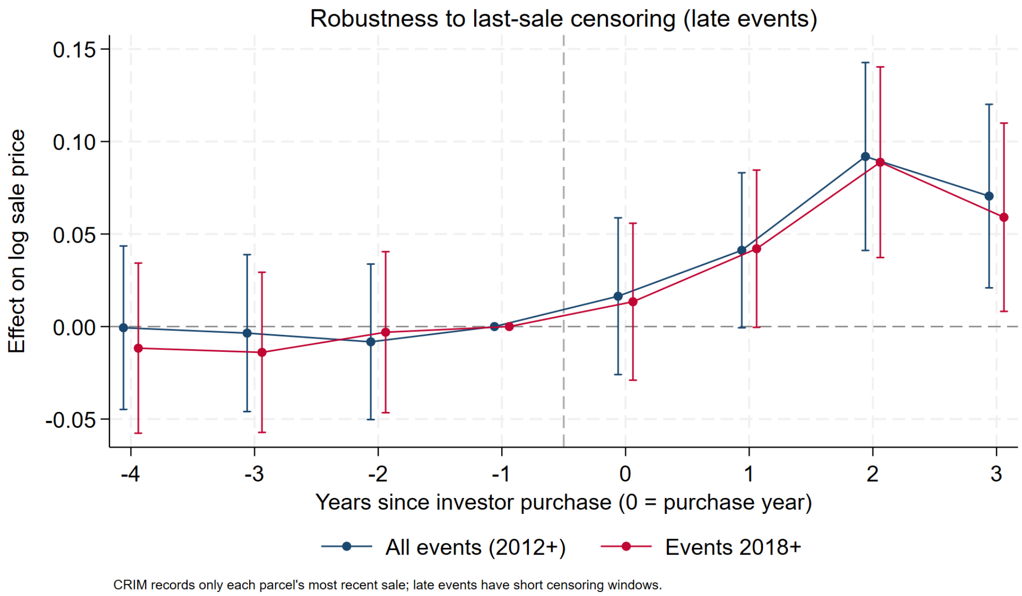

Figure 6: **Censoring robustness.** Baseline overlaid with events from 2018 onward, for which little time exists for post-period sales to be overwritten by later resales. Agreement with the full sample indicates that CRIM's last-sale-only censoring is not manufacturing the bump.

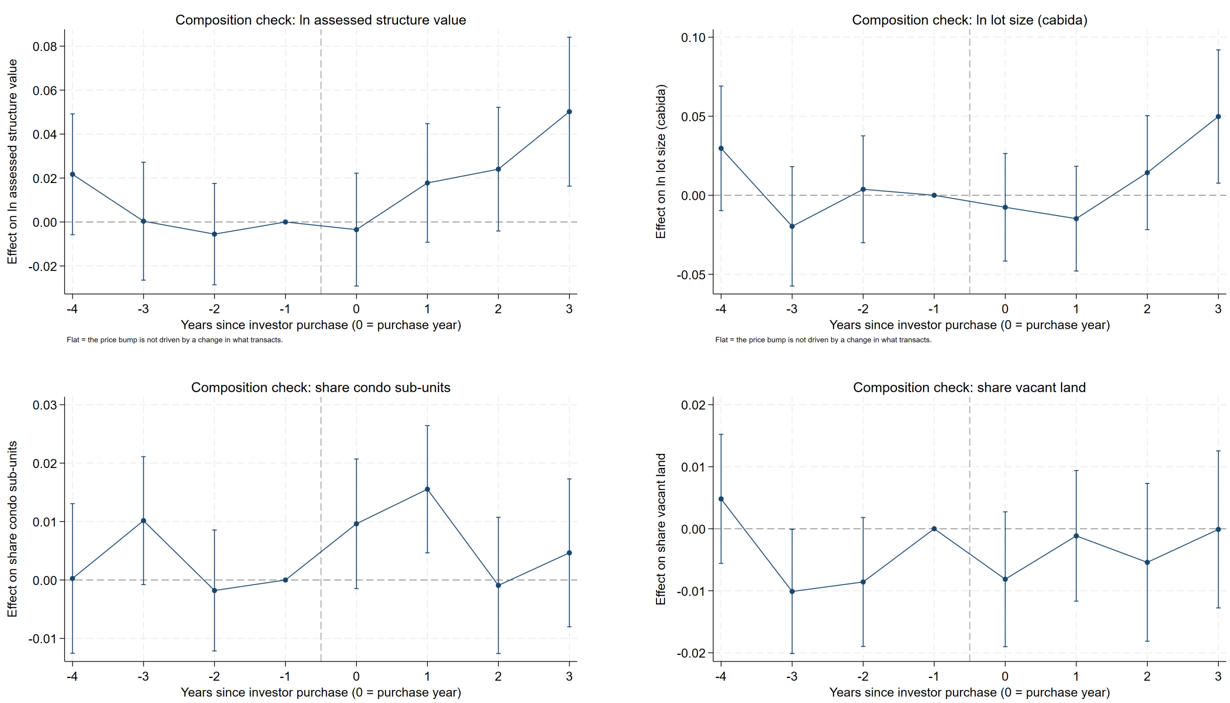

Figure 7: **Composition checks.** The identical design with property characteristics as outcomes: log assessed structure value (top left), log lot size (top right), condo-unit share (bottom left), vacant-land share (bottom right). Flat lines imply the price bump is appreciation on a stable transaction mix; movements would reveal the stock-upgrading margin of gentrification, changing the interpretation of Figure 2.

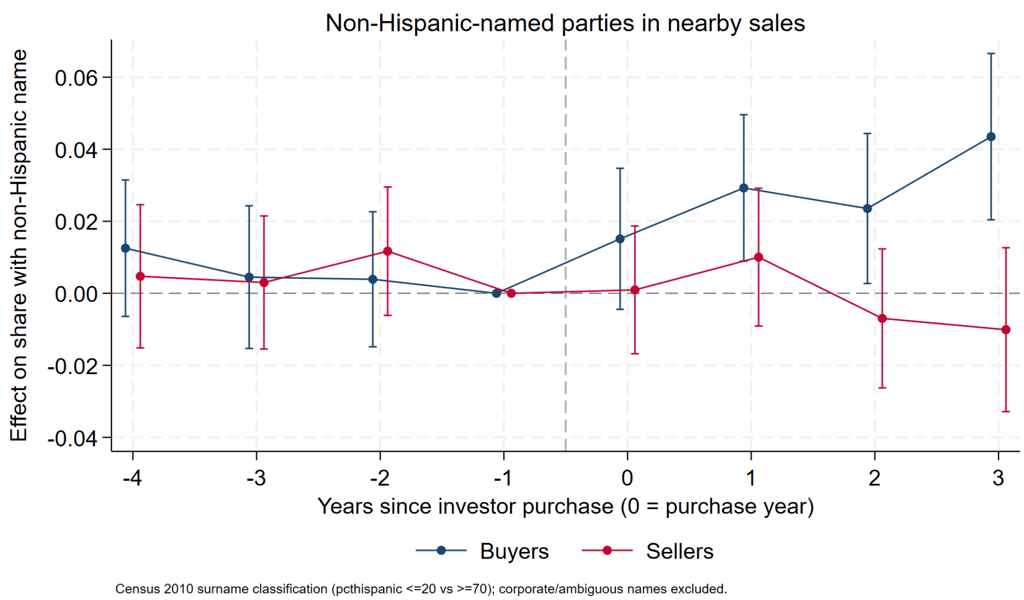

Figure 8: **Who transacts: buyers vs. sellers.** Effects on the share of nearby sales with non-Hispanic-named buyers and, separately, sellers (Census 2010 surname classification; corporate and ambiguous names excluded). Buyer share rising faster than seller share means the area is net-importing mainland owners after an investor arrives — the composition shift price indices cannot see.

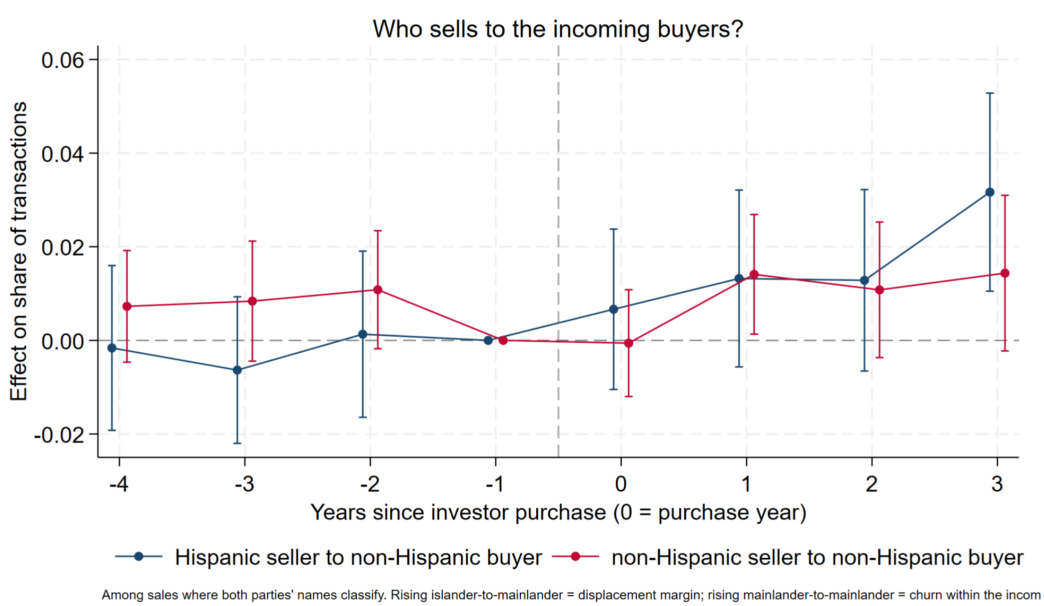

Figure 9: **Transition types.** Effects on the shares of transactions that are Hispanic-seller→non-Hispanic-buyer versus non-Hispanic-seller→non-Hispanic-buyer (among sales where both parties classify). Rising islander→mainlander transitions indicate conversion of locally held housing to incomers (the displacement margin); rising mainlander→mainlander indicates churn within an already-converted segment — together distinguishing "investors lead conversion" from "investors cluster where conversion already occurred."

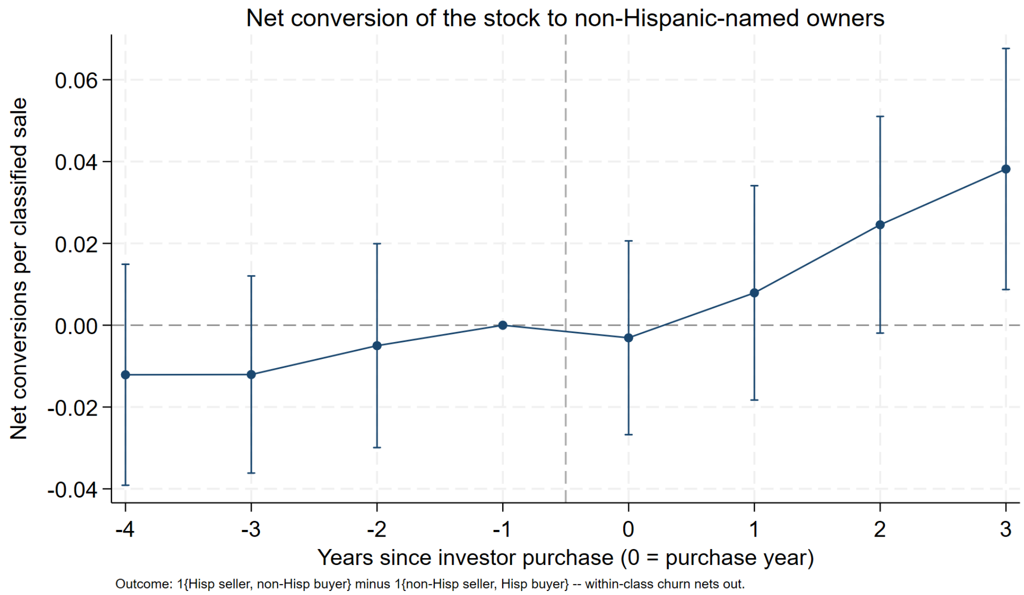

Figure 10: **Net compositional inflow.** The two transition series combined into one displacement measure: 1{Hispanic seller, non-Hispanic buyer} minus 1{non-Hispanic seller, Hispanic buyer} per classified sale. A sale changes the ownership composition of the stock only when the parties' classes differ, so within-class churn (including mainlander→mainlander resales) nets out; a positive post-event effect is net conversion of the nearby stock from Hispanic-named to non-Hispanic-named owners. Pre-period flat (-0.5pp, s.e. 1.0); the net conversion rate rises to $+2.5$pp and $+3.8$pp per sale in years 2–3; post$-$pre $+2.4$pp (s.e. 0.9, $p < 0.01$) per classified sale.

5 The tract-level design and reconciliation (Design 2)

5.1 Data and specification

The tract-level design asks what the ring design structurally cannot: whether investor purchases move *whole neighborhoods*, including the component common to treated areas that near–far contrasts difference out. The outcome panel is the Red Atlas tract$\times$month transaction file (918 of 981 2020-vintage tracts, 2008–2025), aggregated to tract$\times$year with the annual price defined as the transaction-count-weighted mean of monthly tract means (composition-noise reduction; the simple mean is a robustness variant). Treatment timing comes from the dated decree-era events: a tract is treated from the year of its first identified investor purchase (235 treated tracts; 1,200 events), with never-treated tracts as controls. Estimation is annual LP-DiD (pre-window 4, post-window 3) on $100 \times \ln(\text{price})$, with and without county fixed effects.

Table 2 compares treated and never-treated tracts on 2010 PRCS characteristics, all predetermined relative to Act 22. Treated tracts are richer (median income +\$5,700), higher-value (+\$39,000), more educated, less poor, and markedly more seasonal-home- and mainlander-oriented — exactly the profile of investor selection into coastal-amenity areas. Residualizing out municipio fixed effects shrinks the gaps only modestly (15–30%; all remain significant): treated tracts are distinct *within* municipios, not just across them. Selection on 2010 observables is therefore a fact to be handled, not assumed away — which is what Table 3 does, interacting the full 2010 covariate vector with the boom era on top of county$\times$year fixed effects.

Table 2: **2010 Characteristics by Tract Treatment Status** This table presents summary statistics for 2010 PRCS (ACS 2006–2010 5-year) tract characteristics — all pre-determined relative to Act 22 (January 2012) — separated by tracts that ever receive an identified decree-era investor purchase (treated) and tracts that never do. Column (7) reports the raw mean difference between treated and never-treated tracts and column (8) the difference after residualizing out municipio (county) fixed effects, with heteroskedasticity-robust p-values in parentheses. Dollar variables in 2010 dollars; shares in [0,1]. 2010 tracts are mapped to 2020 boundaries by largest land overlap, population-weighted. Statistical significance levels are indicated as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	Treated Tracts			Never-Treated Tracts			Mean Diff.	Adj. Mean Diff.
							[(1) – (4)]	[(1) – (4)]
	Mean	Std. Dev.	Observations	Mean	Std. Dev.	Observations	p-value	p-value
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Median household income ($)	24,522	13,955	218	18,821	9,039	657	5,701***	4,774***
							(0.000)	(0.000)
Median home value ($)	148,938	81,270	218	109,762	36,943	656	39,176***	28,126***
							(0.000)	(0.000)
Median gross rent ($)	515	206	216	442	169	647	73***	62***
							(0.000)	(0.000)
Poverty rate	0.400	0.169	218	0.476	0.160	659	-0.076***	-0.065***
							(0.000)	(0.000)
BA+ share (25+)	0.270	0.168	218	0.188	0.104	660	0.082***	0.071***
							(0.000)	(0.000)
Renter share	0.286	0.155	218	0.292	0.156	659	-0.006	-0.035***
							(0.594)	(0.003)
Vacancy rate	0.196	0.119	218	0.152	0.072	659	0.044***	0.037***
							(0.000)	(0.000)
Seasonal-home share	0.058	0.102	218	0.024	0.032	659	0.034***	0.030***
							(0.000)	(0.000)
Born-in-mainland share	0.057	0.032	218	0.047	0.022	660	0.010***	0.007***
							(0.000)	(0.002)
Population	4,549	1,935	218	3,859	2,004	718	690***	972***
							(0.000)	(0.000)

5.2 Results

Prices in treated tracts rise by a pooled +7.6 log points (s.e. 1.8) after the first investor purchase, with flat pre-period coefficients; adding county fixed effects leaves the estimate unchanged (+7.5), and the simple-mean construction gives +6.1 (Figure 11). Splitting treated tracts by investor concentration reproduces the dose pattern at tract scale: tracts with five or more identified purchases (44 tracts) show +18.3 (2.9), tracts with one to four show +3.8 (2.1) (Figure 12). Transaction *volume* falls in treated tracts, but its pre-period coefficients are significantly negative, so parallel trends fails for that outcome and we treat it as descriptive only.

Table 3 subjects the tract-level estimates — prices, borrower income, and purchase counts by ethnicity — to a common specification ladder, all estimated with the paper's baseline pre-mean-differenced LP-DiD: (1) calendar-year effects, (2) county×year effects, and (3) additionally the ten standardized 2010 PRCS baseline characteristics interacted with the boom-era Post indicator. In column (3) the interactions turn on at the unit's own treatment date, so its Treated coefficient is the effect at sample-mean 2010 characteristics, with the interactions absorbing effect heterogeneity along observables. The price effect moves from 6.4 to 5.7 to 3.9 ($p < 0.01$, $p < 0.01$, $p < 0.05$); borrower income is essentially untouched (8.1, 7.8, 7.4, all $p < 0.01$); Hispanic purchase counts stay a precise null; and non-Hispanic purchase entry (100×asinh) goes 12.7, 12.2, then 3.6 (n.s.) at mean characteristics — the entry margin is genuinely concentrated in high-amenity tracts rather than uniform. Table 4 runs the same ladder on the ring design's cell panel (specification (1) reproduces the baseline estimates exactly): the price effect goes 5.3, 4.2, 1.9 (n.s. at mean characteristics — consistent with the dose results, the ring price effect lives in high-concentration areas), while the net ownership-conversion rate is fully robust: 2.4, 2.5, 2.8pp (all $p < 0.01$). County-year shocks move no estimate much; the column-(3) attenuation of the price and entry effects reflects their concentration in high-amenity, high-dose areas rather than confounding, as the dose tables show directly. Table 5 collects the concentration results of both designs on the common 1–4/5+ dose convention, and Table 6 reports the placebo contrast in table form.

Table 3: **Tract-level treatment effects are robust to county-year shocks and baseline-trend controls** This table reports Treated coefficients from the paper's baseline estimator — the pre-mean-differenced LP-DiD (Dube et al. 2023; H=3, k=4, clean-control sample: newly treated tracts at onset plus never-treated tracts) — under three specifications per outcome: (1) calendar-year effects; (2) county×year effects; (3) adds ten standardized 2010 PRCS baseline characteristics (median household income, home value, and gross rent; poverty, BA+, renter, vacancy, seasonal-home, and mainland-born shares; log population) interacted with the unit's post-treatment indicator (which turns on at the tract's first identified purchase; in the differenced clean sample this is $X\times$ treatment entry), so the column (3) Treated coefficient is the effect at sample-mean 2010 characteristics. Tract fixed effects are subsumed by the long-differencing. Panel A: tract log price (Red Atlas, count-weighted annual mean) and log mean purchase-borrower income (HMDA, 2012–2024), both scaled by 100. Panel B: purchase origination counts by borrower ethnicity as 100 × asinh (Poisson does not compose with pre-mean differencing; the Poisson event studies appear in the figures). Robust standard errors are clustered at the tract level and reported in parentheses. Statistical significance levels are indicated as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Panel A: prices and borrower income

	100 × Log(Price)			*100 × Log(Borrower Income)*		
	(1)	(2)	(3)	(1)	(2)	(3)
Treated	6.44***	5.66***	3.91**	8.10***	7.81***	7.37***
	(1.41)	(1.59)	(1.99)	(1.99)	(2.23)	(2.56)
Observations	11,714	11,681	11,047	8,319	8,299	7,998
R-squared	0.153	0.257	0.265	0.054	0.160	0.172
Tract differencing (LP-DiD)	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-year effects	Yes	No	No	Yes	No	No
County-year effects	No	Yes	Yes	No	Yes	Yes
2010 Controls × Post-Treat.	No	No	Yes	No	No	Yes

Panel B: purchase originations by borrower ethnicity (100×asinh)

	Hispanic Purchases			*Non-Hispanic Purchases*		
	(1)	(2)	(3)	(1)	(2)	(3)
Treated	4.04	2.97	3.17	12.75***	12.16***	3.59
	(5.36)	(5.75)	(4.71)	(3.00)	(3.09)	(2.56)
Observations	9,414	9,392	8,124	9,414	9,392	8,124
R-squared	0.088	0.167	0.191	0.019	0.111	0.146
Tract differencing (LP-DiD)	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-year effects	Yes	No	No	Yes	No	No
County-year effects	No	Yes	Yes	No	Yes	Yes
2010 Controls × Post-Treat.	No	No	Yes	No	No	Yes

Table 4: **Ring-design treatment effects are robust to county-year shocks and baseline-trend controls** This table subjects the ring design's estimates to the same specification ladder as the tract-level outcomes, using the paper's baseline estimator throughout: the pre-mean-differenced LP-DiD on the event×ring cell panel (H=3, k=4; clean-control sample: newly treated near cells at onset plus never-treated cells; specification (1) reproduces the baseline table exactly). Specifications: (1) calendar-year effects; (2) county×year effects (the county is the event's municipio); (3) adds ten standardized 2010 PRCS characteristics of the event's tract interacted with the cell's post-treatment indicator ($X\times$ treatment entry in the differenced clean sample), so the column (3) Treated coefficient is the effect at sample-mean 2010 characteristics. Cell fixed effects are subsumed by the long-differencing. Outcomes: cell mean log sale price (×100) and the net Hispanic-to-non-Hispanic ownership conversion rate per classified sale (percentage points). Robust standard errors are clustered at the cell level and reported in parentheses. Statistical significance levels are indicated as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	100 × Log(Price)			*Net H→NH Conversion (pp)*		
	(1)	(2)	(3)	(1)	(2)	(3)
Treated	5.32***	4.23***	1.91	2.37***	2.47***	2.83***
	(1.40)	(1.34)	(1.45)	(0.88)	(0.86)	(1.00)
Observations	12,379	12,122	11,088	12,035	11,772	10,752
R-squared	0.154	0.343	0.351	0.047	0.168	0.176
Cell differencing (LP-DiD)	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-year effects	Yes	No	No	Yes	No	No
County-year effects	No	Yes	Yes	No	Yes	Yes
2010 Controls × Post-Treat.	No	No	Yes	No	No	Yes

Table 5: **Price Effects by Investor Concentration: Ring and Tract Designs** This table reports LP-DiD price effects by investor concentration under both designs, on the tract design's dose convention. Columns (1)-(2): ring design, events split by total identified purchases within 1km of the event (the event itself included); each group's near rings (0–250m) vs never-treated control rings, cell-level annual LP-DiD. Columns (3)-(4): tract design, treated tracts split by identified purchases in the tract, each group vs never-treated tracts. The Treated coefficient is the post−pre effect from the single pre-mean-differenced LP-DiD regression (Dube et al. 2023), whose SE, R-squared, and Observations are reported. All outcomes 100×log price; SEs clustered by cell (ring) or tract. Statistical significance levels are indicated as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	Ring design		*Tract design*	
	1–4 purch.	5+ purch.	1–4 purch.	5+ purch.
	(1)	(2)	(3)	(4)
Treated	-2.74	6.41***	4.13***	16.30***
	(3.45)	(1.50)	(1.59)	(2.54)
Observations	2,854	9,525	11,670	11,476
R-squared	0.074	0.217	0.153	0.152
Unit differencing (LP-DiD)	Yes	Yes	Yes	Yes
Calendar-year effects	Yes	Yes	Yes	Yes
Never-treated controls	Yes	Yes	Yes	Yes

Table 6: **Investor Purchases vs Placebo Luxury Purchases** Placebo events are 1,500 purchases by individual non-investor buyers in the investor price band, drawn from the same micro-areas and more than three years from any real event; both series use far rings matched at 400–750m. A price response unique to investor purchases rules out generic luxury-transaction effects. The Treated coefficient is the post−pre effect from a single LP-DiD regression via pre-mean differencing (Dube, Girardi, Jordà and Taylor 2023): dependent variable $\frac{1}{4}\sum_{h=0}^{3} y_{t+h} - \frac{1}{4}\sum_{\tau=t-4}^{t-1} y_\tau$ on treatment entry with calendar-time effects, clean-control sample; its SE, R^2 and Observations come from that regression. Coefficients are 100×log points (shares in pp). SEs clustered by unit (cell/tract). Statistical significance levels are indicated as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	Investor purchases	Placebo purchases
	(1)	(2)
Treated	4.43***	-0.33
	(1.46)	(1.40)
Observations	12,036	7,932
R-squared	0.109	0.258
Calendar-year effects	Yes	Yes
Event×ring cell differencing	Yes	Yes
Hedonic controls	No	No

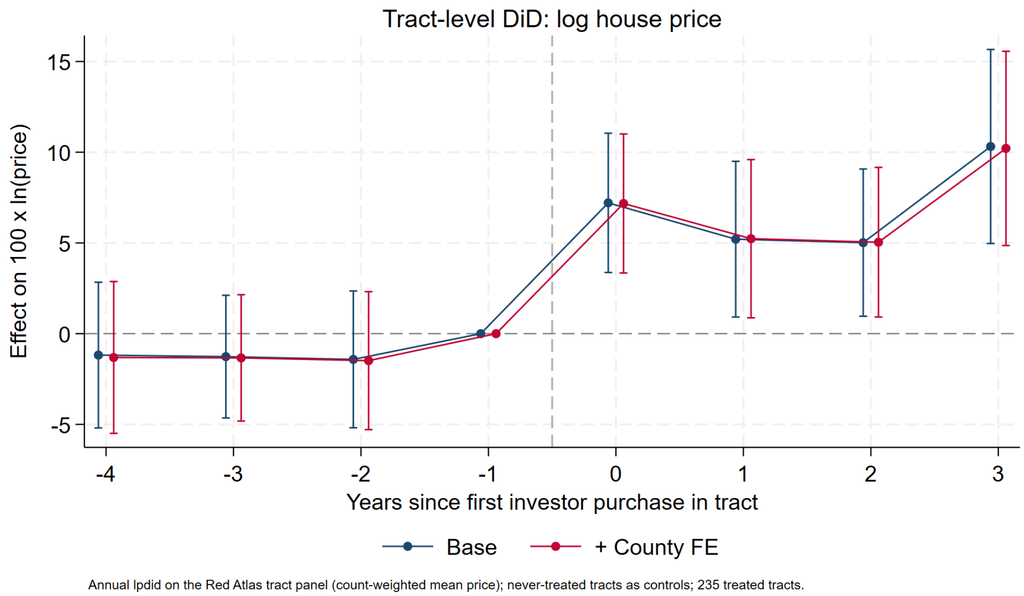

Figure 11: **Tract-level event study: log house price.** Annual LP-DiD on the Red Atlas panel, 235 treated tracts vs never-treated controls; base and county-FE series are indistinguishable and pre-trends are flat.

5.3 Reconciliation: the whole vs the sum of the parts

Because the ring design delivers a distance gradient, it implies a *prediction* for each treated tract: the stock-value-weighted average of $\beta(d)$ over the tract's parcels — the tract-level effect if micro-spillovers were the whole story. Comparing observed tract-DiD estimates to these predictions decomposes the aggregate (Figure 13): tracts with 1–4 purchases sit at or below their prediction (observed +3.8 vs predicted +5.4) — micro-spillovers fully account for them — while the 44 concentration tracts exceed their prediction by roughly 11 log points (observed +18.3 vs predicted +7.5). A market-level channel beyond proximity operates only where investors concentrate.

5.4 Aggregation

Integrating the estimated gradient over the exposure distribution of the island's improved housing stock (60.0% of parcels, 63.1% of assessed value, lie within 2.5 km of an identified event) yields an island-wide average price effect of +2.1 to +3.0%, or \$6.3–8.0B of housing value at band-specific market/assessed ratios — roughly 6–8% of Puerto Rico's post-2019 boom (FHFA purchase-only index: +64% 2019–2024). Adding the concentration excess in the 44 high-dose tracts (≈\$4.3B) raises the identified program footprint to ≈\$11–12B, about 10–11% of the boom. At municipio level, using municipio repeat-sales indices as the local boom denominators, footprints in the top investor municipios are much larger (Table 7).

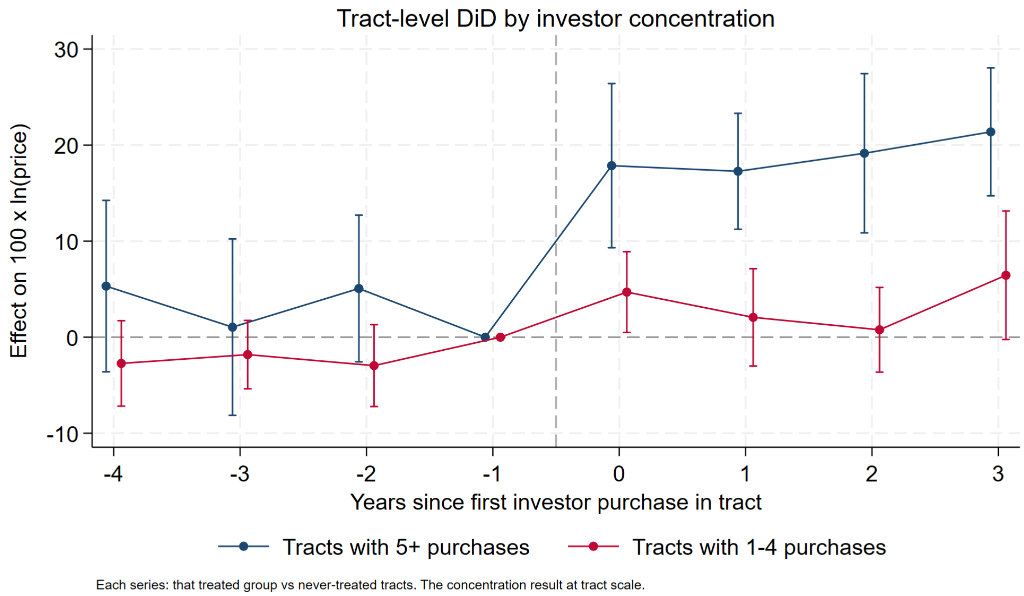

Figure 12: **Tract-level event study by investor concentration.** Tracts with 5+ identified purchases versus tracts with 1–4, each against never-treated controls: the concentration result at tract scale.

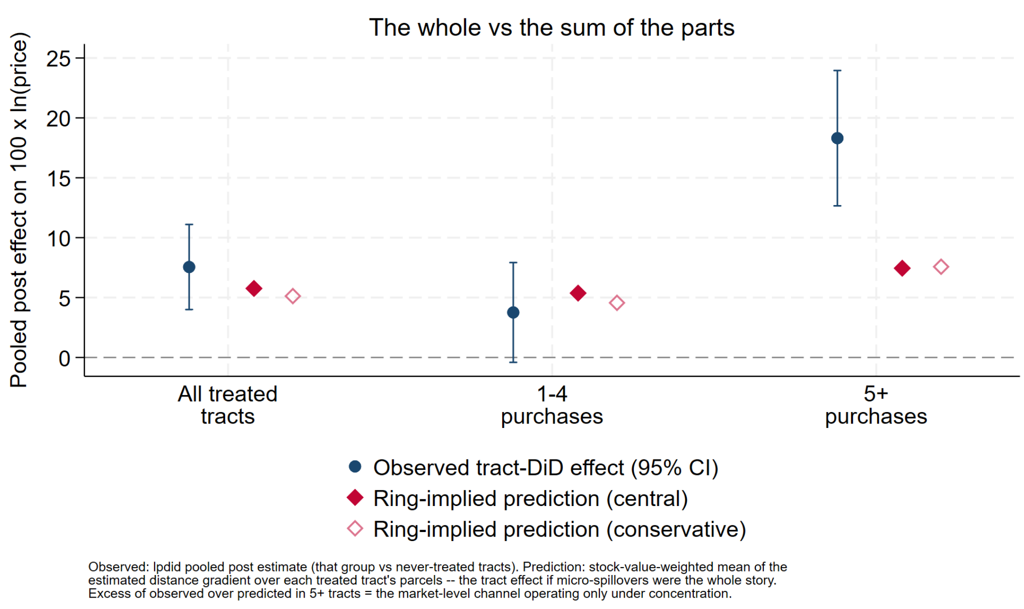

Figure 13: **Observed tract effects vs ring-implied predictions.** Observed: LP-DiD pooled post estimates by dose group. Predictions: stock-value-weighted distance-gradient effects over each treated tract's parcels. The all-treated and 1–4-purchase groups sit on their predictions; the 5+ group's excess (≈11 log points) is the concentration channel.

Table 7: **Program Footprint in the Most-Exposed Municipios** This table reports each municipio's post-2019 housing boom and the share of it attributable to identified investor purchases under three accountings. All columns share one valuation convention: each parcel at assessed value times its tract's median market/assessed ratio (2019+ sales; municipio and island fallbacks). The dollar boom is the municipio's market stock times $1 - 1/(1+g)$, the portion of current value attributable to repeat-sales growth g since 2019; Uplift is the synthesis in dollars. Shares of boom connect exactly: *Ring, Total* integrates the distance gradient over all parcels and *5+ tracts* restricts to parcels in 5+-purchase tracts; *Tract DiD* applies the dose-specific tract effects to treated-tract stock (*Total*: all treated tracts; *5+ tracts*: the high-dose subset). The synthesis *5+ differential* is Tract DiD (5+) minus Ring (5+) — the tract-wide repricing beyond what proximity predicts — and *Synthesis, Total* = Ring (Total) + the 5+ differential, so the ring layer is never double-counted.

	Local boom '19–'24			Synthesis		Ring design		Tract DiD	
Municipio	Growth	$	Uplift	Total	5+ diff.	Total	5+ tracts	Total	5+ tracts
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Rincón	+148%	$1.1B	$0.34B	31.1%	17.7%	13.4%	12.7%	30.7%	30.4%
Dorado	+177%	$3.7B	$0.97B	25.9%	14.4%	11.5%	10.3%	25.0%	24.8%
San Juan	+89%	$17.8B	$4.29B	24.1%	9.0%	15.1%	6.0%	17.8%	15.1%
Río Grande	+115%	$2.0B	$0.48B	24.0%	15.7%	8.4%	6.1%	23.5%	21.8%
Carolina	+91%	$7.6B	$1.32B	17.2%	6.8%	10.4%	4.7%	12.2%	11.6%
Guaynabo	+124%	$5.4B	$0.84B	15.5%	2.3%	13.2%	1.7%	8.6%	4.0%
Humacao	+121%	$2.5B	$0.39B	15.5%	7.2%	8.3%	5.1%	13.7%	12.4%

All aggregates are lower bounds: exposure counts only the 1,200 name-identified purchases, excluding LLC and unmatched investors; the capitalization step applies transaction-price effects to the full stock; and event effects predating 2019 enter the numerator while the boom clock starts in 2019.

5.5 Mortgage originations: volumes are a convincing null

HMDA loan-level data for Puerto Rico (2018–2024; 105,559 originations, aggregated to 2020-vintage tract×year) show no detectable response of mortgage lending in treated tracts: Poisson (PPML) estimates with tract and year fixed effects are statistically zero for owner-occupied purchases (-0.5%, s.e. 5.4), non-owner-occupied purchases (+3.0%, 11.3), rate/term and cash-out refinancings, and total originations, and are unchanged with county×year fixed effects. Consistent with decree investors transacting predominantly in cash, the program's effects appear in prices and buyer composition rather than local credit expansion.

5.6 Who can still buy: borrower composition in treated tracts

Splitting purchase originations by HMDA's self-reported borrower ethnicity (Hispanic or Latino vs. not; joint and unavailable excluded) sharpens the displacement interpretation. Since decree investors buy in cash, HMDA is effectively a census of the *financed* — overwhelmingly local, 94.6% Hispanic — side of the market. To get adequate pre-periods we extend the panel back to 2012 using

CFPB's historic HMDA files, on a population that is consistent across the 2017/2018 reporting change: originated first-lien, owner-occupied, 1–4 family home-purchase loans (2012–2024; annual volumes are smooth across the seam). Three results (Figures 14–15). First, borrower *income* rises after treatment — pooled +9.6% (s.e. 1.9), reaching +10.0% at $h \geq 3$ — and with up to six pre-years of data (displayed with the endpoint binned at -4) the pre-period is now flat (-3.7, -1.1, -2.4, individually and jointly insignificant, with no run-up into treatment; the apparent pre-drift in the short 2018–2024 panel was an artifact of its truncated window). Second, the income shift is concentrated where investors concentrate: +17.6% (s.e. 4.9) in 5+-purchase tracts vs. +7.9% (s.e. 2.0) in 1–4-purchase tracts. Third, on the extensive margin, Hispanic-borrower purchase counts are unchanged (pooled +0.6%, s.e. 4.5) while *non-Hispanic* borrower purchases rise +37% (s.e. 14) from a thin base — financed mainlander entry is now detectable, not just cash entry. Together: local buyers as a group keep transacting, but the marginal financed buyer moves up the income distribution, most strongly in high-dose tracts — the affordability-displacement margin, measured on self-reported ethnicity and income rather than the surname proxy.

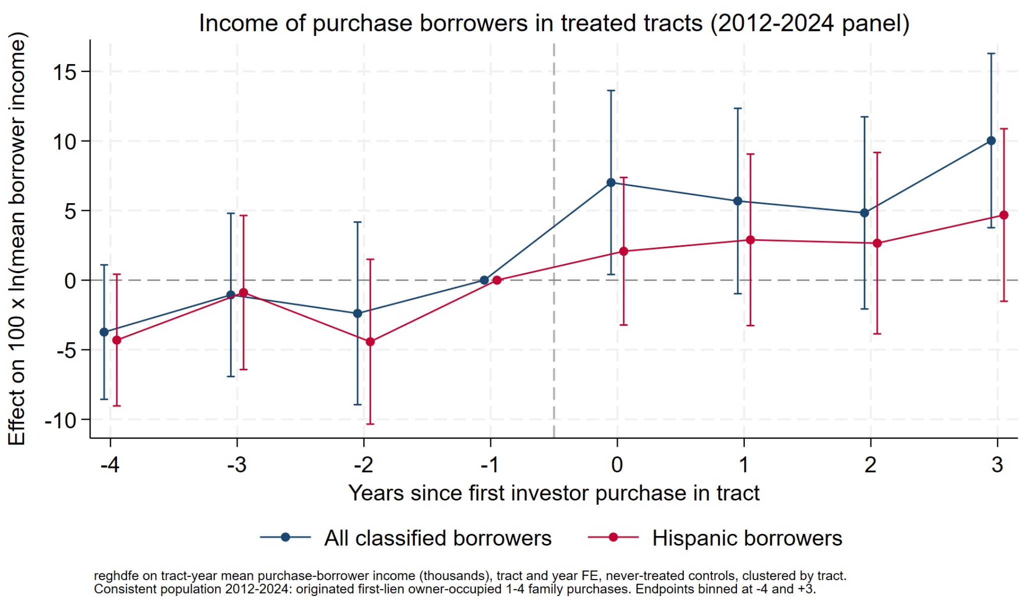

Figure 14: **Income of purchase borrowers, 2012–2024 panel.** Effect on $100 \times \ln$(mean borrower income) among first-lien owner-occupied purchase originations in treated tracts; all classified borrowers and Hispanic borrowers only. No pre-trend (endpoint binned at -4 pools horizons back to -6); pooled +9.6% (s.e. 1.9).

Financed mainlander entry

The non-Hispanic borrower series make the entry margin visible on its own terms (Figures 16–17; thin base — a few hundred non-Hispanic-borrower loans per year island-wide — hence the wide intervals). Purchase originations by non-Hispanic borrowers rise +37% (s.e. 14) pooled and +41% (s.e. 19) at $h \geq 3$ in treated tracts, with noisy but non-trending pre-periods: decree-era in-movers are

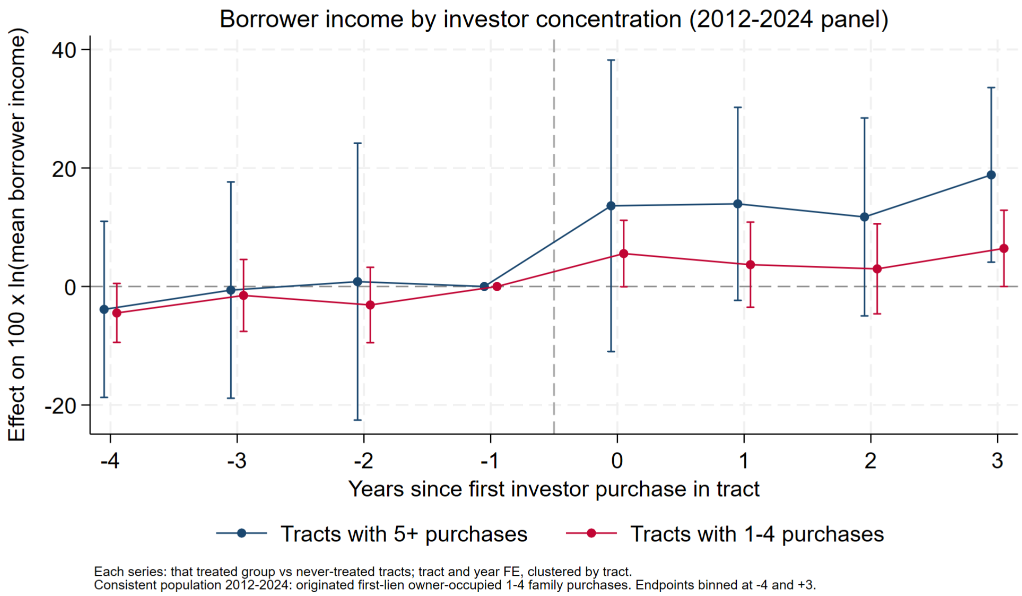

Figure 15: **Borrower income by investor concentration, 2012–2024 panel.** Each series: that treated group vs. never-treated tracts. The income shift is twice as large in 5+-purchase tracts (+17.6%, s.e. 4.9) as in 1–4-purchase tracts (+7.9%, s.e. 2.0) — the same concentration gradient as prices.

not exclusively cash buyers — a financed segment follows the same geography.[1] Their refinancings rise even more steeply — +70% (s.e. 17) pooled, +176% (s.e. 34) at $h \geq 3$ — consistent with newcomers refinancing recently acquired properties as the areas appreciate. (The corresponding *Hispanic* refinancing series is not usable as evidence: its pre-period climbs steadily into treatment with no post-treatment kink, so the incumbent equity-access margin is unidentified and we do not report it.)

6 Standing caveats

1. **Last-sale censoring** (CRIM): pre-period sales are survivors that never resold; addressed by design and Figure 5.

2. **Site selection**: investors choose appreciating micro-areas; estimates are local spillovers *conditional on* location choice, not "the effect of the Act."

[1] This magnitude reconciles with the ring design's buyer-composition effect once the two bases are compared. The CRIM non-Hispanic-*named* buyer share rises +2.5pp (s.e. 0.7, PMD post−pre) on a pre-treatment near-ring mean of 18.9% — nominally +13%. But the surname proxy inflates the base: self-reported non-Hispanic borrowers are only 1.6% of financed purchases in treated tracts pre-treatment, and non-Hispanic-*named* buyers are 10% of the general market against 67% among confirmed investors, so much of the 18.9pp proxy base is Hispanic-heritage locals with non-Hispanic-classifying surnames — a component inert to treatment. If the treatment-responsive (true mainlander) share of the proxy base is roughly a third of it (6–7pp), the +2.5pp effect is +35–40% of the responsive base, matching the HMDA estimate almost exactly. Both sources thus indicate mainlander purchasing rose by roughly one-third of its pre-treatment level — CRIM measuring all transactions at ring geography, HMDA the financed tail at tract geography.

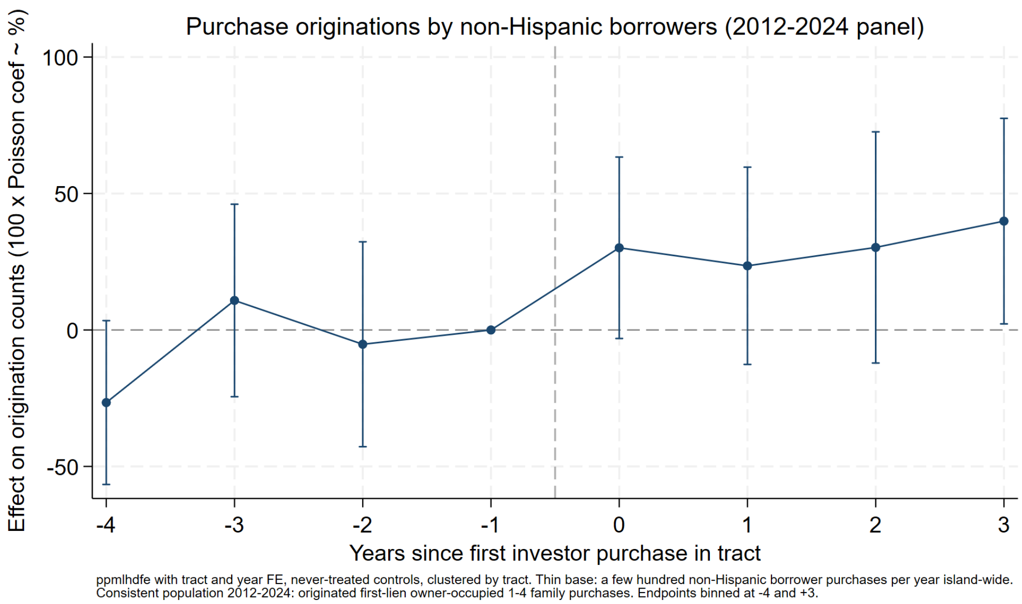

Figure 16: **Purchase originations by non-Hispanic borrowers, 2012–2024 panel.** Poisson event study, treated vs. never-treated tracts. Pooled +37% (s.e. 14).

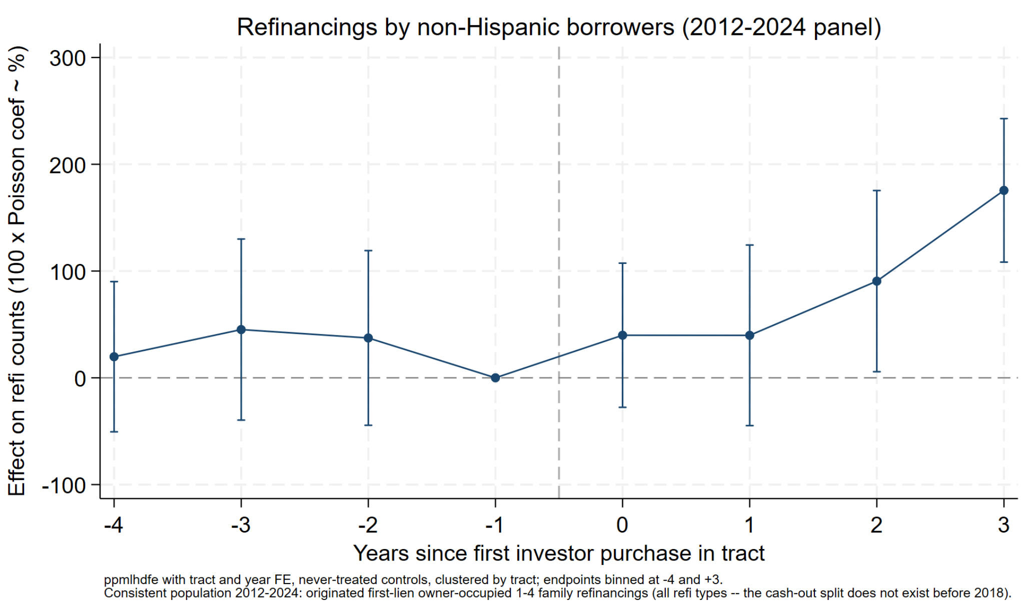

Figure 17: **Refinancings by non-Hispanic borrowers, 2012–2024 panel.** Same population (first-lien owner-occupied 1–4 family; all refi types — the cash-out split does not exist before 2018). Pooled +70% (s.e. 17), rising to +176% (s.e. 34) at $h \geq 3$.

3. **Match selection**: identified investors skew toward single-property, personal-name buyers; entity (LLC/trust) purchases are missing.

4. **Name proxy**: understates mainland presence; corporate and ambiguous names are excluded from shares.

5. **Assessed values** are on CRIM's 1957 basis — valid as cross-sectional controls and composition outcomes, never as market values.